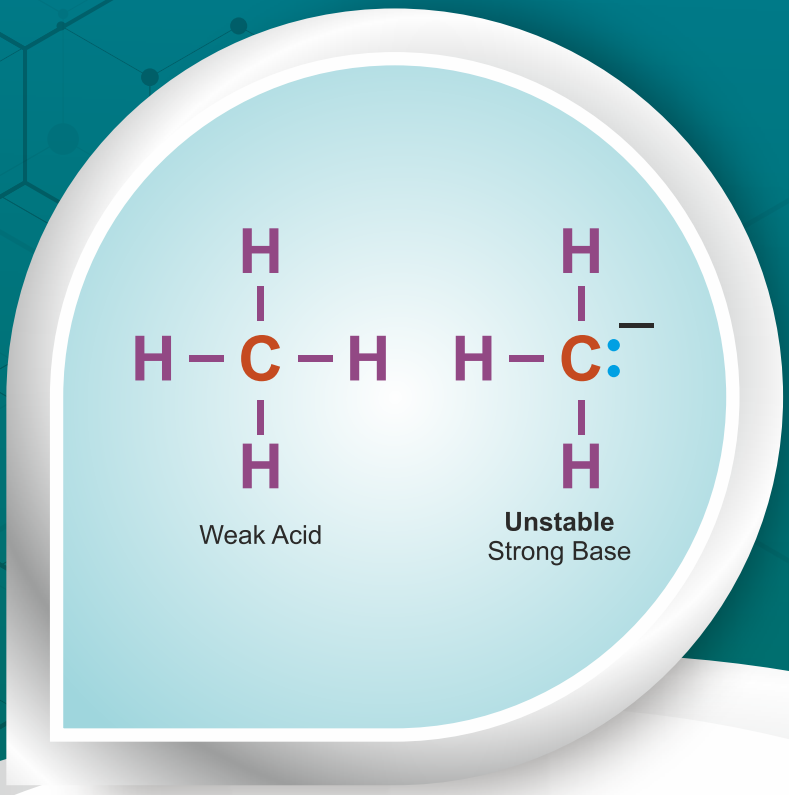
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ORGANIC CHEMISTRY

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NURTURE COURSE



Exercise

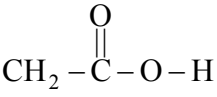
Acidic Strength & Basic Strength

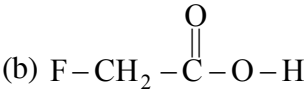
(English Medium)

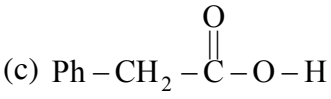
ACID STRENGTH & BASIC STRENGTH

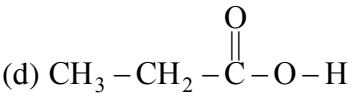
EXERCISE # O-I

1. Write correct order of acidic strength of following compounds :

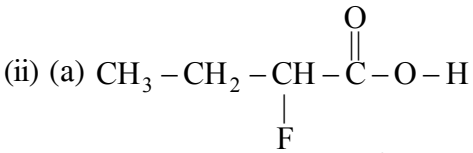
(i) (a) 

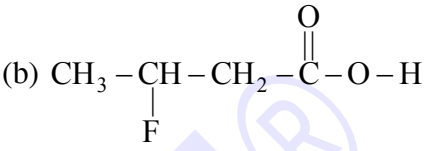
(b) 

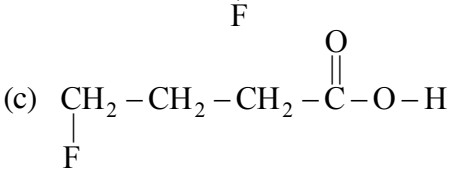
(c) 

(d) 

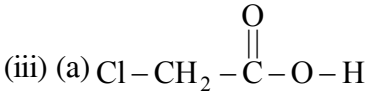
AB0001

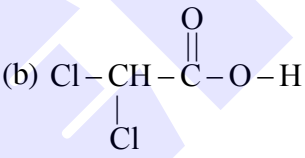
(ii) (a) 

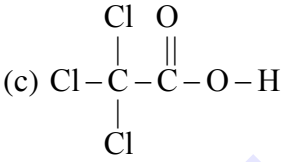
(b) 

(c) 

AB0002

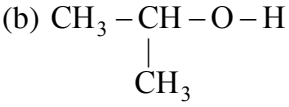
(iii) (a) 

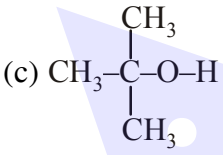
(b) 

(c) 

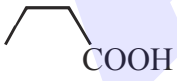
AB0003

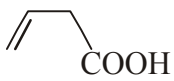
(iv)(a) $CH_3–CH_2–O–H$

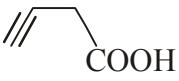
(b) 

(c) 

AB0004

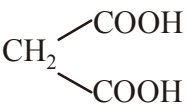
(v) (a) 

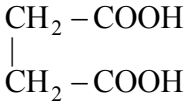
(b) 

(c) 

AB0005

(vi) (a) 

(b) 

(c) 

AB0006

(vii) (a) H–F (b) H–Cl (c) H–Br (d) H–I

AB0007

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(viii) (a) CH_4 (b) NH_3 (c) H_2O (d) H–F

AB0008

(ix)(a) $F–CH_2–CH_2–O–H$ (b) $NO_2–CH_2–CH_2–O–H$

(c) $Br–CH_2–CH_2–O–H$ (d) $\overset{\oplus}{N}H_3–CH_2–CH_2–O–H$

AB0009

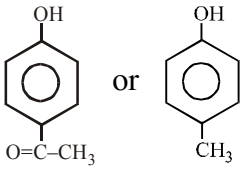
(x) (a) CH_3COOH (b) CH_3CH_2OH (c) C_6H_5OH (d) $C_6H_5SO_3H$

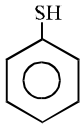
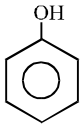
AB0010

2. Explain which is a stronger acid.

(a) CH_3CH_3 or $BrCH_2NO_2$ **AB0011**

(b) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$ or $CH_3-\overset{\overset{O}{\|}}{C}-CH_2CN$ **AB0012**

(c)  or **AB0013**

(d)  or  **AB0014**

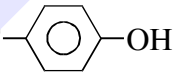
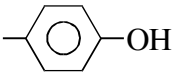
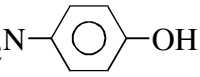
3. Which of the following would you predict to be the stronger acid ?

(a) Benzoic acid or para-nitrobenzoic acid **AB0015**

(b) $CH_3–CH_2–CH_2–OH$ or $CH_3–CH = CH – OH$ **AB0016**

(c) $CH_3 – CH = CH – CH_2 – OH$ or $CH_3 – CH = CH – OH$ **AB0017**

4. Arrange the given phenol & its derivative in their decreasing order of acidity :

(I) $C_6H_5–OH$ (II)  (III)  (IV) 

Select the correct answer from the given code:

(A) IV > III > I > II (B) IV > II > III > I (C) IV > III > II > I (D) IV > I > III > II

AB0018

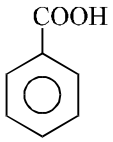
5. Which one of the following is the most acidic?

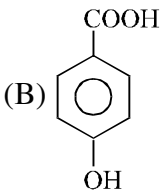
(A)  (B)  (C)  (D) $CH_2=CH–CH_3$

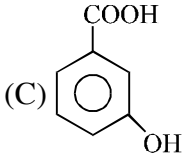
AB0019

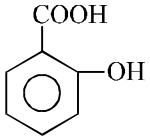
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6. Which of the following is weakest acid?

(A) 

(B) 

(C) 

(D) 

AB0020

7. Arrange pH of the given compounds in decreasing order:

(1) Phenol (2) Ethyl alcohol (3) Formic acid (4) Benzoic acid

(A) 1 > 2 > 3 > 4 (B) 2 > 1 > 4 > 3 (C) 3 > 2 > 4 > 1 (D) 4 > 3 > 1 > 2

AB0021

8. Arrange acidity of given compounds in decreasing order:

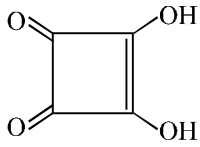
(I) $CH_3–NH–CH_2–CH_2–OH$ (II) $CH_3–NH–CH_2–CH_2–CH_2–OH$

(III) $(CH_3)_3\overset{\oplus}{N}-CH_2-CH_2-OH$

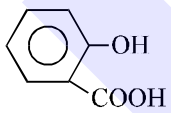
(A) III > I > II (B) III > II > I (C) I > II > III (D) II > I > III

AB0022

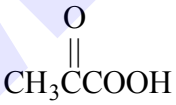
9. Consider the following compound



I



II



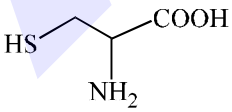
III

Which of the above compounds reacts with $NaHCO_3$ giving CO_2

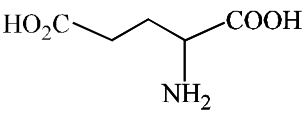
(A) I, II and III (B) I and III (C) II and III (D) I and II

AB0023

10. Say which pk_a belong to which functional group in case of following amino acids :

(i) cysteine : 

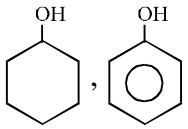
1.8, 8.3 & 10.8 **AB0024**

(ii) glutamic acid : 

: 2.19, 4.25, 9.67 **AB0025**

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11. Record the following sets of compounds according to increasing pK_a (= – log Ka)

(a)  , cyclohexane carboxylic acid. **AB0026**

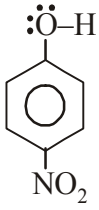
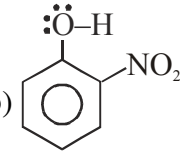
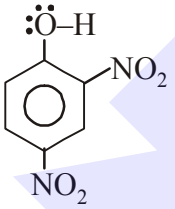
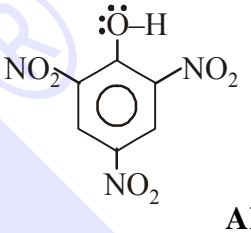
(b) 1-butyne, 1-butene, butane **AB0027**

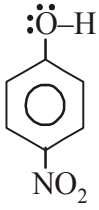
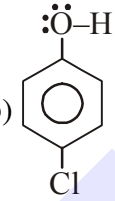
(c) Propanoic acid, 3-bromopropanoic acid, 2-nitropropanoic acid **AB0028**

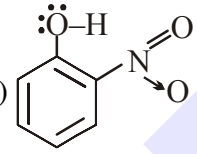
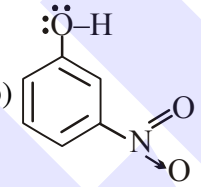
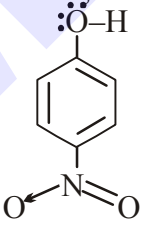
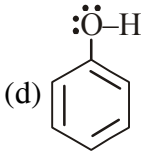
(d) Phenol,o-nitrophenol, o-cresol **AB0029**

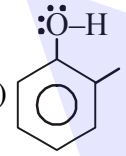
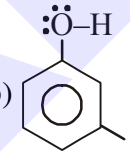
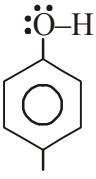
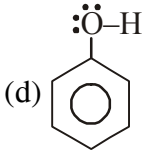
(e) Hexylamine, aniline, methylamine **AB0030**

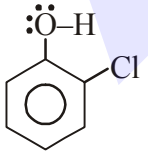
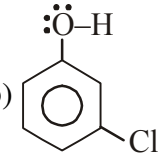
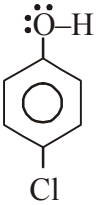
12. Write correct order of acidic strength of following compounds:

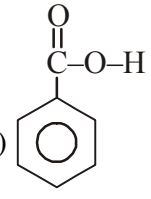
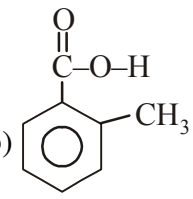
(i) (a)  (b)  (c)  (d)  **AB0031**

(ii) (a)  (b)  (c)  **AB0032**

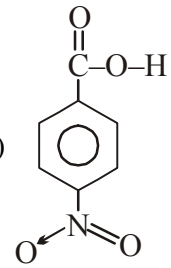
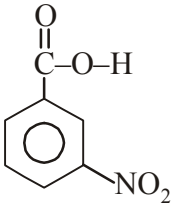
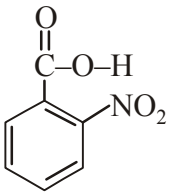
(iii) (a)  (b)  (c)  (d)  **AB0033**

(iv) (a)  (b)  (c)  (d)  **AB0034**

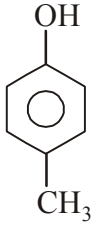
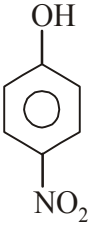
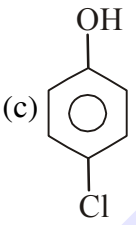
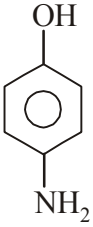
(v) (a)  (b)  (c)  **AB0035**

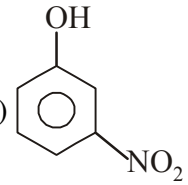
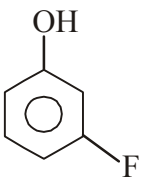
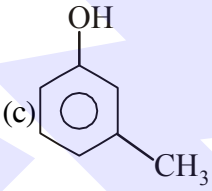
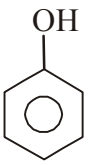
(vi) (a)  (b)  **AB0036**

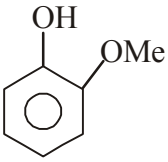
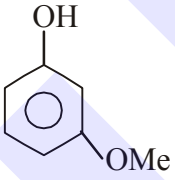
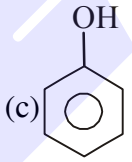
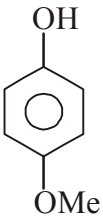
node06808C-BD\Kota\JEE(Advanced)\Nurture\Chem Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength \01_Theory & Ex..p65

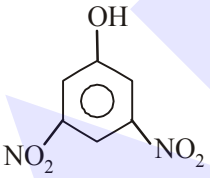
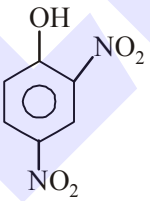
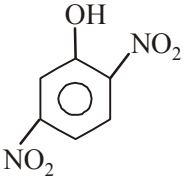
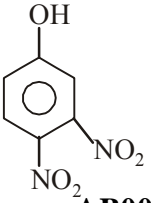
(vii) (a)  (b)  (c)  **AB0037**

13. Select the strongest acid in each of the following sets :

(i) (a)  (b)  (c)  (d)  **AB0038**

(ii) (a)  (b)  (c)  (d)  **AB0039**

(iii) (a)  (b)  (c)  (d)  **AB0040**

(iv) (a)  (b)  (c)  (d)  **AB0041**

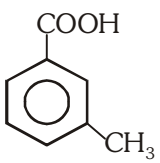
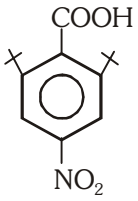
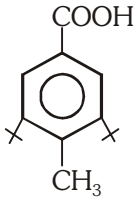
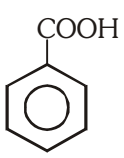
14. The strongest acid is :

(A)HF (B) CH_3CO_2H (C) $HF + SbF_5$ (D) H_2S **AB0042**

15. The weakest acid (does not show acidic character) is :

(A)$HC \equiv CH$ (B) $CH_2 = CH_2$ (C) Me_3CH (D) Ph_3CH **AB0043**

16. Which of the following is most acidic :

(A)  (B)  (C)  (D)  **AB0044**

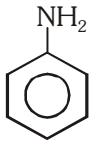
node06808C-BD\Kota\JEE\Advanced\Nurture\Chem\Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength\01_Theory & Ex_p65

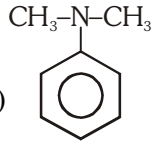
Paragraph for Question 17 to 18

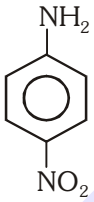
The most important condition for resonance to occur is that the involved atoms in resonating structure must be coplanar or nearly coplanar for maximum delocalisation. If this condition does not fulfil, involved orbitals cannot be parallel to each other and as a consequence delocalisation cannot occcur. Bulky groups present on adjacent atoms inhibit the planarity of atoms involved in resonance. This phenomenon is known as steric inhibition of resonance. Steric inhibition of resonance has profound effect on

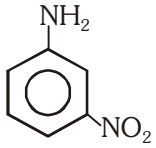
(1) Physical properties (2) Acidity and basicity (3) Reactivity of organic compounds

17. Arrange the following in the decreasing order of basicity :

(I) 

(II) 

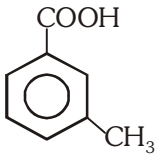
(III) 

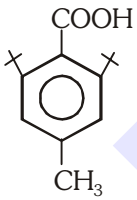
(IV) 

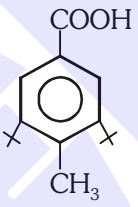
(A) I > II > III > IV (B) IV > III > II > I (C) II > I > IV > III (D) I > IV > III > II

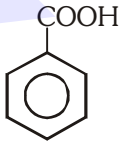
AB0045

18. Which of the following is most acidic :

(A) 

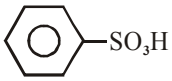
(B) 

(C) 

(D) 

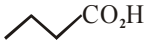
AB0046

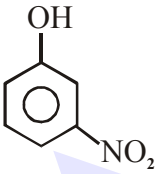
19. How many following compounds are more acidic than water ?

(a) 

(b) HCl

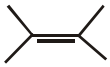
(c) $CH_3 - C \equiv CH$

(d) 

(e) 

(f) 

(g) 

(h) 

(i) NaOH

AB0047

20. Select correct order regarding acidic strength of given compounds :

(1) o-methylbenzoic acid (2) m-methylbenzoic acid

(3) p-methylbenzoic acid (4) benzoic acid

(A) 1 > 2 > 3 > 4 (B) 4 > 3 > 2 > 1 (C) 1 > 4 > 2 > 3 (D) 3 > 2 > 4 > 1

AB0048

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EXERCISE # O-II

1. Write decreasing order of basic strength of following :

(i) (a) CH_3^- (b) NH_2^- (c) OH^- (d) F^-

AB0049

(ii) (a) F^- (b) Cl^- (c) Br^- (d) I^-

AB0050

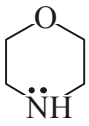
(iii) (a) NH_3 (b) $MeNH_2$ (c) Me_2NH (d) Me_3N (in H_2O)

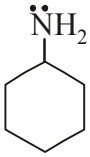
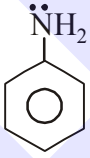
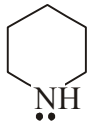
AB0051

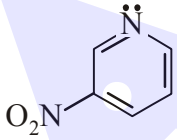
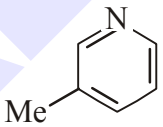
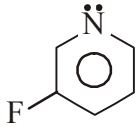
(iv) (a) NH_3 (b) $MeNH_2$ (c) Me_2NH (d) Me_3N (Gas phase)

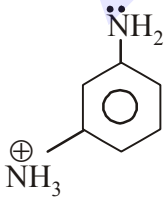
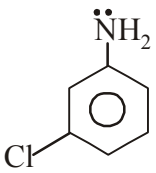
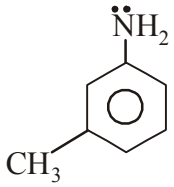
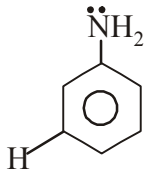
AB0052

(v) (a) $R–NH_2$ (b) $Ph–NH_2$ (c) $R-\overset{\overset{O}{\|}}{C}-NH_2$ **AB0053**

(vi) (a)  (b)  (c)  **AB0054**

(vii)(a)  (b)  (c)  **AB0055**

(viii) (a)  (b)  (c)  **AB0056**

(ix) (a)  (b)  (c)  (d) 

AB0057

node06808C-BD\Kota\JEE\Advanced\Nurture\Chem\Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength\01_Theory & Ex_p65

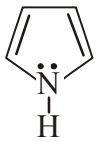
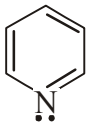
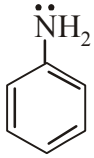
2. Write decreasing order of basic strength of following :

(i) (a) $CH_3–CH_2–\ddot{N}H_2$ (b) $CH_3–CH=\ddot{N}H$ (c) $CH_3–C\equiv\ddot{N}$

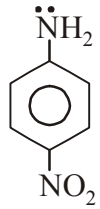
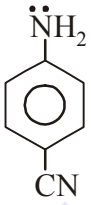
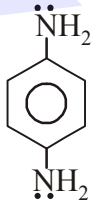
AB0058

(ii) (a) $CH_3-\underset{\underset{O}{\|}}{C}-\ddot{N}H_2$ (b) $CH_3–CH_2–\ddot{N}H_2$ (c) $CH_3-\underset{\underset{\ddot{N}H}{\|}}{C}-\ddot{N}H_2$ (d) $\ddot{N}H_2-\underset{\underset{\ddot{N}H}{\|}}{C}-\ddot{N}H_2$

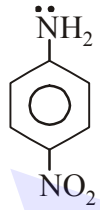
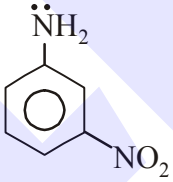
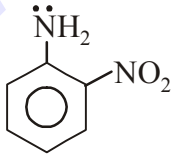
AB0059

(iii) (a)  (b)  (c) 

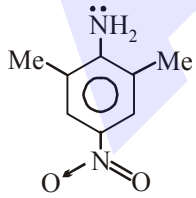
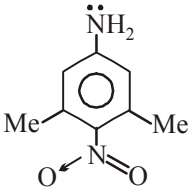
AB0060

(iv) (a)  (b)  (c)  (d) 

AB0061

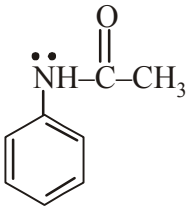
(v) (a)  (b)  (c) 

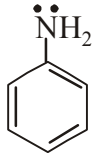
AB0062

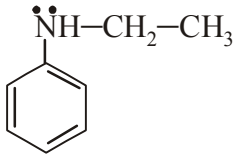
(vi) (a)  (b) 

AB0063

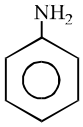
node06808C-BD\Kota\JEE[Advanced]\Nurture\Chem Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength \01_Theory & Ex..p65

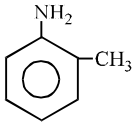
(vii)(a) 

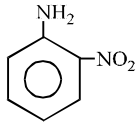
(b) 

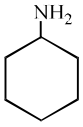
(c) 

AB0064

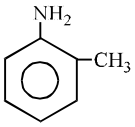
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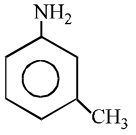
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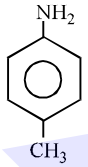
(c) 

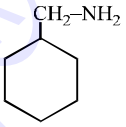
(d) 

AB0065

(ix) (a) 

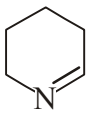
(b) 

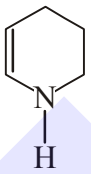
(c) 

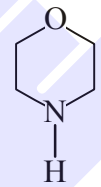
(d) 

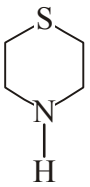
AB0066

3. Select the strongest base in following compound :

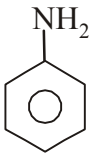
(i) (a) 

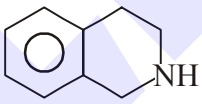
(b) 

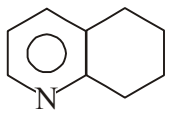
(c) 

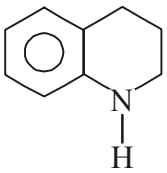
(d) 

AB0067

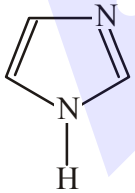
(ii) (a) 

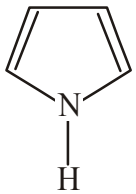
(b) 

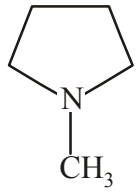
(c) 

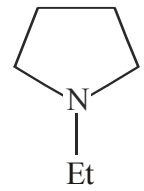
(d) 

AB0068

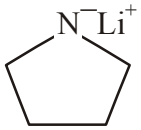
(iii) (a) 

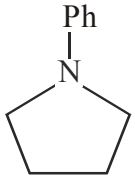
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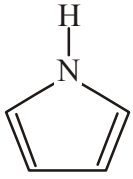
(c) 

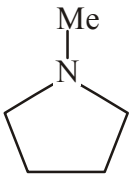
(d) 

AB0069

(iv) (a) 

(b) 

(c) 

(d) 

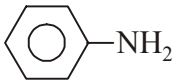
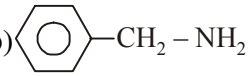
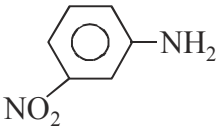
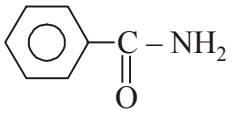
AB0070

node06808C-BDXdata\JEE\Advanced\Nurture\Chem\Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &c B Strength\01_Theory & Ex_p65

4. Arrange the following compound in decreasing order of their basicity.

(i) (a) $H_2C = CHNa$ (b) CH_3CH_2Na (c) CH_3CH_2ONa (d) $HC \equiv CNa$

AB0071

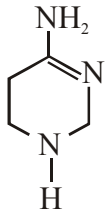
(ii) (a)  (b)  (c)  (d) 

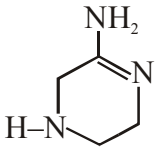
AB0072

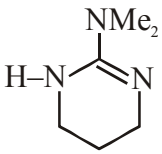
(iii) (a) HO^- (b) NH_3 (c) H_2O (d) HSO_4^-

AB0073

5. Correct decreasing order of basic strength -


(I)


(II)


(III)

Of following compound -

(A) III > II > I (B) II > I > III (C) I > II > III (D) III > I > II

AB0074

6. Consider the following bases:

(I) o-nitroaniline (II) m-nitroaniline (III) p-nitroaniline

The decreasing order of basicity is:

(A) II > III > I (B) II > I > III (C) I > II >III (D) I > III > II

AB0075

7. Consider the basicity of the following aromatic amines:

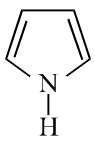
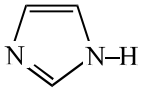
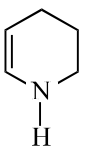
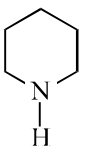
(I) aniline (II) p-nitroaniline (III) p-methoxyaniline (IV) p-methylaniline

The correct order of decreasing basicity is:

(A) III > IV > I > II (B) III > IV > II > I (C) I > II > III > IV (D) IV > III > II > I

AB0076

8. Which one of the following is least basic in character?

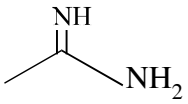
(A)  (B)  (C)  (D) 

AB0077

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9. In each of the following pair of compounds, which is more basic in aqueous solution? Give an explanation for your choice:

(a) CH_3NH_2 or CF_3NH_2 **AB0078**

(b) CH_3CONH_2 or H_2N  **AB0078**

(c) $CH_3CH_2CH_2NH_2$ or CH_3CN **AB0078**

(d) $C_6H_5N(CH_3)_2$ or 2,6-dimethyl-N-N-dimethylaniline **AB0078**

10. Choose the member of each of the following pairs of compunds that is likely to be the weaker base.

(a) H_2O or H_3O^+ **AB0079**

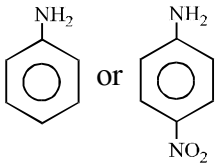
(b) Cl^-, SH^- **AB0079**

(c) F^-, OH^-, NH_2^-, CH_3^- **AB0079**

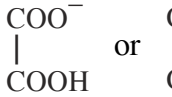
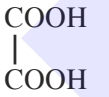
(d) HF, H_2O, NH_3 **AB0079**

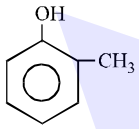
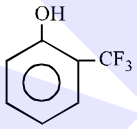
(e) OH^-, SH^-, SeH^- **AB0079**

11. Explain which compound is the weaker base.

(a)  or **AB0080**

(b) $CH_2 = CH - CH = CH - CH_2^-$ or $CH_2 = CH - CH_2^-$ **AB0080**

(c)  or  **AB0080**

(d)  or  **AB0080**

12. Arrange the basic strength of the following compounds.

(a) OH^- CH_3COO^- Cl^-

(i) (ii) (iii)

AB0081

(b) $CH \equiv C^-$ $CH_2 = CH^-$ $CH_3CH_2^-$

(i) (ii) (iii)

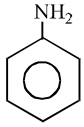
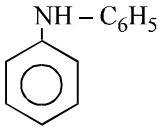
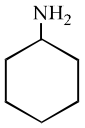
AB0082

(c) $CH_2 = CHCH_2NH_2$ $CH_3CH_2CH_2NH_2$ $CH \equiv C - CH_2NH_2$

(i) (ii) (iii)

AB0083

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(d)   

(i) (ii) (iii)

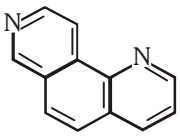
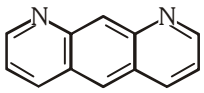
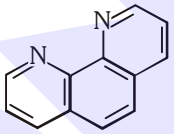
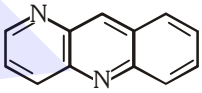
AB0084

13. Arrange the following compounds in order of increasing basicity.

(a) CH_3NH_2, $CH_3NH_3^{\oplus}$, CH_3NH^- **AB0085**

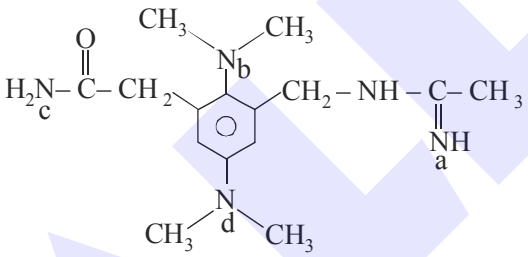
(b) CH_3O^-, CH_3NH^-, $CH_3CH_2^-$ **AB0085**

14. Which of the following is most basic :

(A)  (B)  (C)  (D) 

AB0086

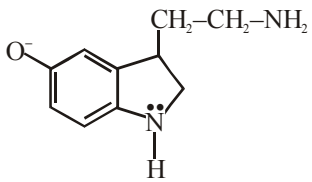
15. Basicity order of N in following compound is :



(A) b > d > a > c (B) a > b > d > c (C) a > b > c > d (D) a > c > b > d

AB0087

16. The conjugate base of serotonin (used as tranquilisers) is given as follows :

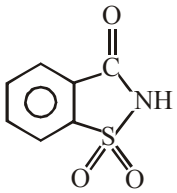


How many basic groups present in following compound ?

AB0088

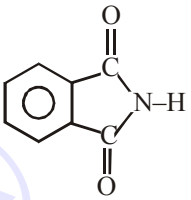
node06808C-BDVKap\JEE(Advanced)\Nurture\Chem Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength \01_Theory & Ex.p65

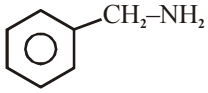
17. The structure of saccharin is given as follows :



[Provided N is the donor atom in all cases]

How many following compounds are more basic than saccharin ?

(i) $CH_3-\underset{\underset{NH}{\|}}{C}-NH_2$ (ii) $H_2N-\underset{\underset{NH}{\|}}{C}-NH_2$ (iii) $CH_3-\underset{\underset{O}{\|}}{C}-NH_2$ (iv) 

(v) 

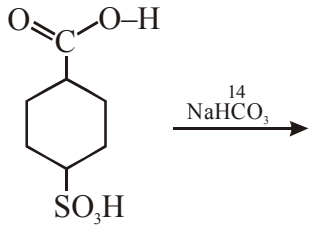
(vi) 

AB0089

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EXERCISE # S-I

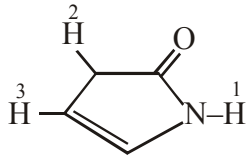
1. In given reaction Gas liberated is/are



(A) CO_2 & SO_3 (B) SO_3 & $^{14}CO_2$ (C) $^{14}CO_2$ only (D) SO_2 only

AB0090

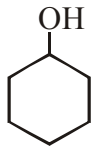
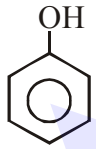
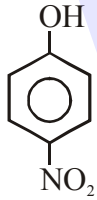
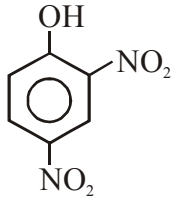
2. Arrange marked atom in decreasing order of acidic strength



(A) 1 > 2 > 3 (B) 3 > 2 > 1 (C) 2 > 1 > 3 (D) 2 > 3 > 1

AB0091

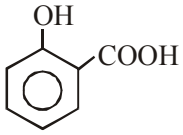
3.

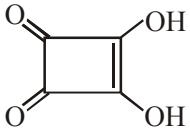
Column - I	**Column-II**
(A) 	(P) React with NaOH
(B) 	(Q) React with $NaHCO_3$
(C) 	(R) React with NaH
(D) 	(S) React with Na
	(T) React with $NaNH_2$

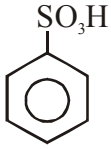
AB0092

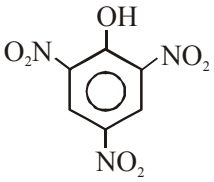
node06808C-BDVKab\JEE[Advanced]\Nurture\Chem Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength \01_Theory & Ex..p65

4. Compound which can give effevescences with $NaHCO_3$

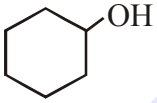
(i)  (Salicylic acid)

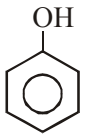
(ii)  (Squaric acid)

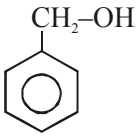
(iii)  (Benzene sulphonic acid)

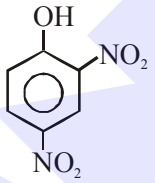
(iv)  (Picric acid)

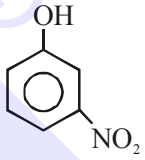
(v) Ph–CH=CH–COOH (cinnamic acid)

(vi) 

(vii) 

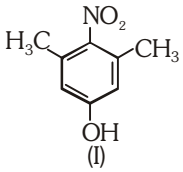
(viii) 

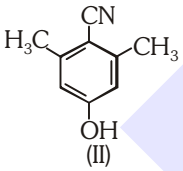
(ix) 

(x) 

AB0093

5. **Statement-1 :** For the given two compounds-I is more acidic than compounds-II.


(I)


(II)

and

Statement-2 : Due to presence of $-CH_3$ group at ortho positions to $-NO_2$; the plane of $-NO_2$ deviates, w.r.t plane of ring.

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.

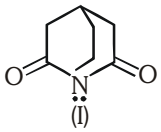
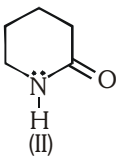
(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

AB0094

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6. **Statement 1** :  is more basic than 

and

Statement 2 : Lone pair electrons on nitrogen in compound (I) does not participate in resonance.

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.

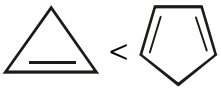
(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

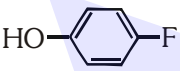
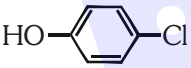
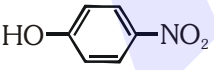
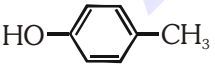
AB0095

7. Match Column-I with Column-II.

Column - I (Facts)	**Column - II (Reasons)**
(A) Guanidine $H_2N-\underset{\underset{NH}{\|}}{C}-NH_2$ is example of strong base	(P) Resonance stabilisation of conjugate acid of strong base.
(B) Carbanion stability $\bar{C}Cl_3 > \bar{C}F_3$	(Q) Due to s-character of central atoms
(C) Alkyne is more acidic than alkene	(R) Due to d-orbital resonance
(D) Acidity : 	(S) Due to formation of aromatic anion
	(T) Stability of conjugate acid / base due to more number of identical resonating structure

AB0096

8. Match Column-I with Column-II.

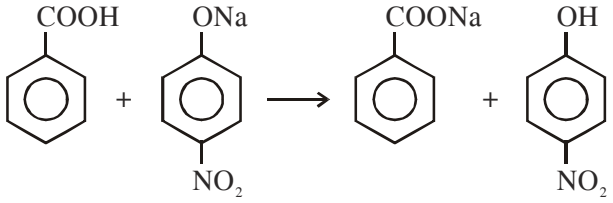
Column - I (Compounds)	**Column - II (pKa)**
(A) 	(P) 7.15
(B) 	(Q) 10.14
(C) 	(R) 9.98
(D) 	(S) 9.38
	(T) pKa is more than phenol

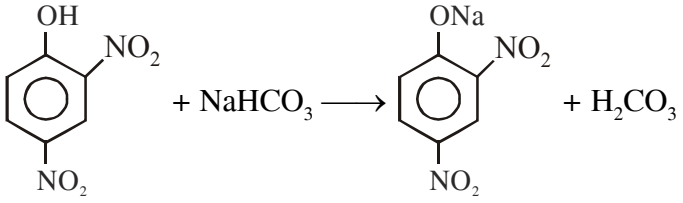
AB0097

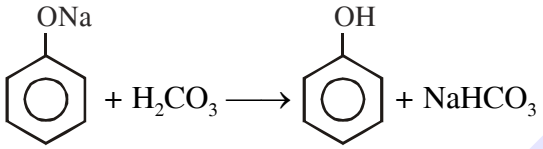
node06808C-BDVKota\JEE(Advanced)\Nurture\Chem Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength \01_Theory & Ex..p65

(Comprehension) (Q.9 to Q.11)

Observe the following feasible reactions :

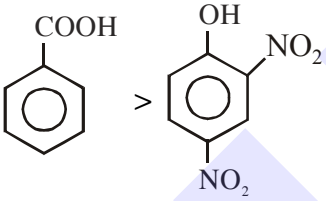
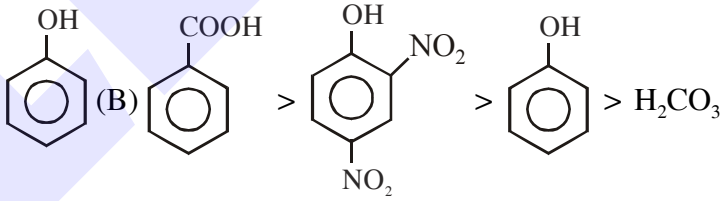
(i) 

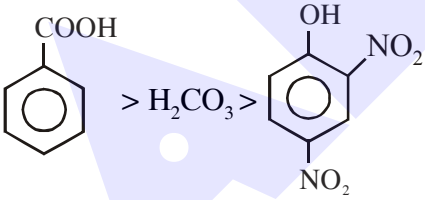
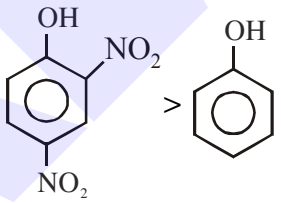
(ii) 

(iii) 

Answer the following question :

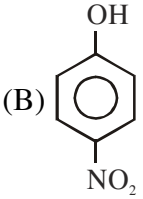
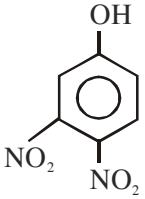
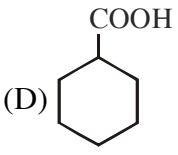
9. Which of the following is the correct order of acidic strength.

(A)  > H_2CO_3 > (B) 

(C)  > H_2CO_3 >  > (D) None

AB0098

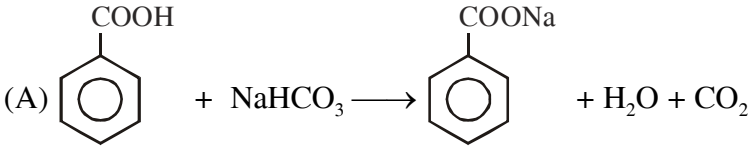
10. Which of the following compound does not react with $NaHCO_3$

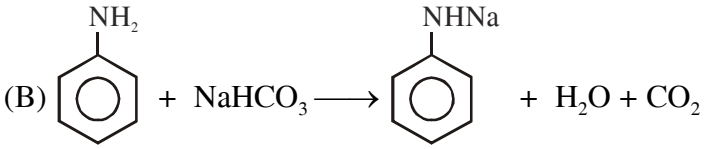
(A) 
(B) 
(C) 
(D) 

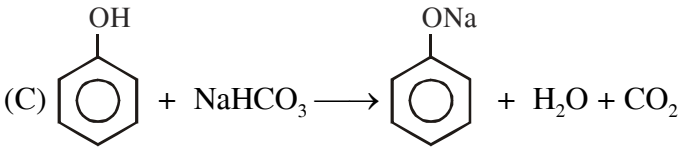
AB0099

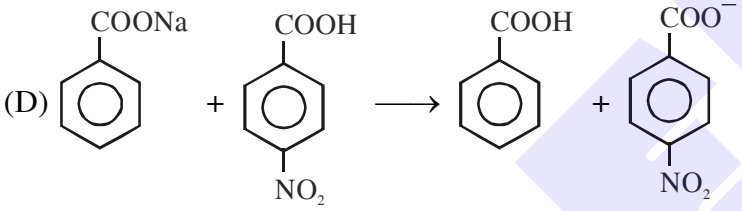
node06808C-BD\Kota\JEE\Advanced\Nurture\Chem\Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength\01_Theory & Ex_p65

11. Identify the feasible reactions

(A) 

(B) 

(C) 

(D) 

AB0100

12. Identify the non-feasible reaction

(A) $CH_3–C \equiv CH + NH_2^- \rightleftharpoons CH_3—C \equiv \overset{\bullet\bullet}{C}^- + NH_3$

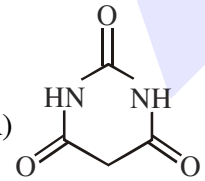
(B) $CH_3CH_2—OH + NaH \rightleftharpoons CH_3CH_2ONa + H_2$

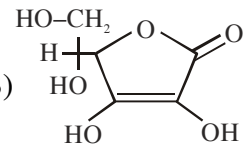
(C) $CH_3—OH + NaOH \rightleftharpoons CH_3ONa + H_2O$

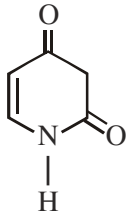
(D) $HC{\equiv}CH + NaOH \rightleftharpoons HC{\equiv}CNa + H_2O$

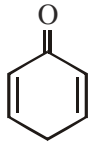
AB0101

13. Select the number of compounds in which deprotonation gives aromatic anion :

(A) 
Barbituric acid

(B) 
Ascorbic acid
(Vitamin-C)

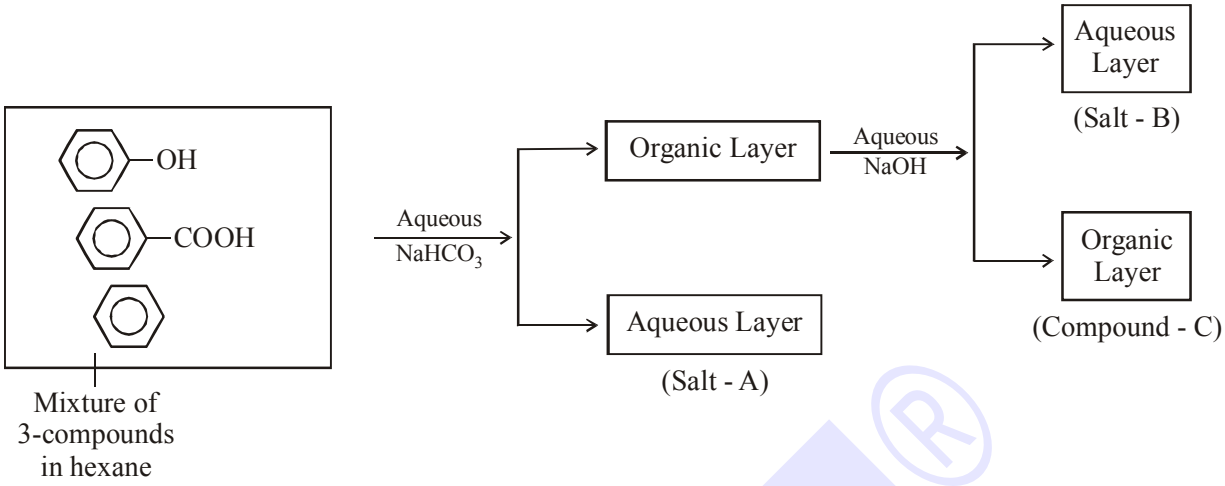
(C) 

(D) 

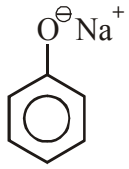
AB0102

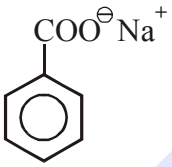
node06808C-BD\Kota\JEE(Advanced)\Nurture\Chem Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &B Strength \01_Theory & Ex..p65

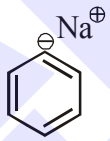
Paragraph for Questions 14 and 15



14. Identify salt 'A' ?

(A) 

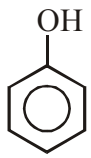
(B) 

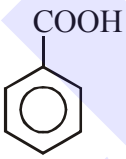
(C) 

(D) All of these

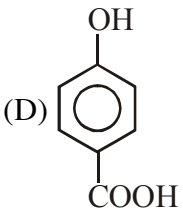
AB0103

15. Identify compound 'C' ?

(A) 

(B) 

(C) 

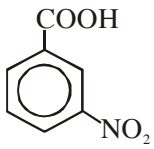
(D) 

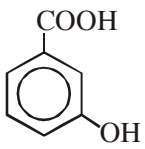
AB0103

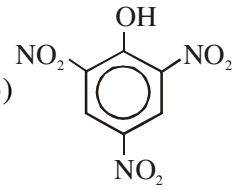
node06808C-BDXcels\JEE\Advanced\Nurture\Chem\Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &c B Strength\01_Theory & Ex_p65

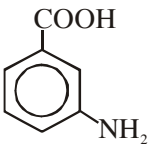
EXERCISE # JEE-MAIN

1. Picric acid is - **[AIEEE-2002]**

(1) 

(2) 

(3) 

(4) 

AB0104

2. Which of the following speices acts both as bronsted acid & base – **[AIEEE-2002]**

(1) NH_3 (2) OH^- (3) $HSO_4^{\ominus}$ (4) 1 and 2 both

AB0105

3. The correct order of increasing basic nature for the bases NH_3, CH_2NH_2 and $(CH_3)_2NH$ is- **[AIEEE-2003]**

(1) $CH_3NH_2 < NH_3 < (CH_3)_2NH$ (2) $(CH_3)_2NH < NH_3 < CH_3NH_2$

(3) $NH_3 < CH_3NH_2 < (CH_3)_2NH$ (4) $CH_3NH_2 < (CH_3)_2NH < NH$

AB0106

4. Consider the acidity of the carboxylic acids- **[AIEEE-2004]**

(i) PhCOOH (ii) o–$NO_2C_6H_4COOH$

(iii) p-$NO_2C_6H_4COOH$ (iv) m-$NO_2C_6H_4COOH$

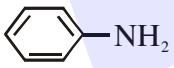
which of the following is the correct order of acidity-

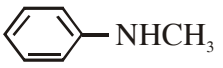
(1) i > ii > iii > iv (2) ii > iv > iii > i

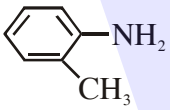
(3) ii > iv > i > iii (4) ii > iii > iv > i

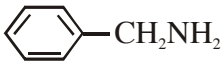
AB0107

5. Which of the following is the strongest base - **[AIEEE-2004]**

(1) 

(2) 

(3) 

(4) 

AB0108

6. Among the following acids which has the lowest pk_a value- **[AIEEE-2005]**

(1) CH_3CH_2COOH (2) $(CH_3)_2CHCOOH$

(3) HCOOH (4) CH_3COOH

AB0109

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7. Amongest the following the most basic compound is- **[AIEEE-2005]**

(1) p-nitro aniline (2) Acetanilide

(3) Aniline (4) Benzylamine

AB0110

8. What is the conjugate base of OH^-? **[AIEEE-2005]**

(1) H_2O (2) O_2 (3) O^{2-} (4) O^-

AB0111

9. Among the following acids which has the lowest pK_a value? **[AIEEE-2005]**

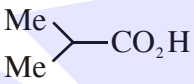
(1) HCOOH (2) CH_3COOH

(3) CH_3CH_2COOH (4) $(CH_3)_2CH–COOH$

AB0112

10. The correct order of increasing acid strength of the compounds is **[AIEEE-2006]**

(a) CH_3CO_2H (b) $MeOCH_2CO_2H$

(c) CF_3CO_2H (d) 

(1) d < a < c < b (2) d < a < b < c

(3) a < d < c < b (4) b < d < a < c

AB0113

11. Which one of the following is the strongest base in aqueous solution ? **[AIEEE-2007]**

(1) Trimethylamine (2) Aniline

(3) Dimethylamine (4) Methylamine

AB0114

12. The correct order of increasing basicity of the given conjugate base (R=CH_3) is :- **[AIEEE-2010]**

(1) $RCO\bar{O} < HC \equiv \bar{C} < \bar{N}H_2 < \bar{R}$ (2) $RCO\bar{O} < HC \equiv \bar{C} < \bar{R} < \bar{N}H_2$

(3) $\bar{R} < HC \equiv \bar{C} < RCO\bar{O} < \bar{N}H_2$ (4) $RCO\bar{O} < \bar{N}H_2 < HC \equiv \bar{C} < \bar{R}$

AB0115

13. The strongest acid amongst the following compounds is ? **[AIEEE-2011]**

(1) $CH_3CH_2CH(Cl)CO_2H$ (2) $ClCH_2CH_2CH_2COOH$

(3) CH_3COOH (4) HCOOH

AB0116

14. The correct order of acid strength of the following compounds :-

A. Phenol B. p-Cresol

C. m-Nitrophenol D. p- Nitrophenol

is :- **[AIEEE-2011]**

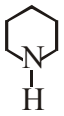
(1) C > B > A > D (2) D > C > A > B

(3) B > D > A > C (4) A > B > D > C

AB0117

node06\B0BC-BD\Kota\JEE\Advanced\Nurture\Chem\Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength\01_Theory & Ex_p65

15. In the following compounds : **[JEE(Main)-2012]**

(I)  (II)  (III)  (IV) 

the order of basicity is as follows :

(1) IV > III > II > I (2) II > III > I > IV

(3) I > III > II > IV (4) III > I > II > IV

AB0118

16. The most basic compound among the following is :- **[JEE(Main)-2012]**

(1) Acetanilide (2) Benzylamine

(3) p-Nitro aniline (4) Aniline

AB0119

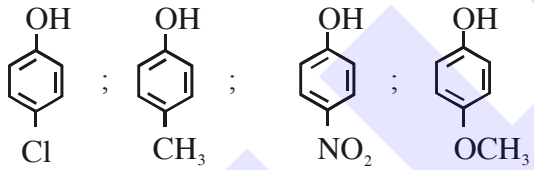
17. The order of basicity of amines in gaseous state is :- **[JEE(Main)-2013]**

(1) $3° > 2° > NH_3 > 1°$ (2) $1° > 2° > 3° > NH_3$

(3) $NH_3 > 1° > 2° > 3°$ (4) $3° > 2° > 1° > NH_3$

AB0120

18. Arrange the following compounds in order of decreasing acidity : **[JEE(Main)-2013]**



(I) (II) (III) (IV)

(1) II > IV > I > III (2) I > II > III > IV

(3) III > I > II > IV (4) IV > III > I > II

AB0121

19. The conjugate base of hydrazoic acid is :- **[JEE(Main)-2014]**

(1) HN_3^- (2) N_3^- (3) N_2^- (4) N^{-3}

AB0122

20. Which one of the following compounds will not be soluble in sodium bicarbonate ?

[JEE(Main)-2014]

(1) Benzene sulphonic acid (2) Benzoic acid

(3) o-Nitrophenol (4) 2, 4, 6 - Trinitrophenol

AB0123

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21. Considering the basic strength of amines in aqueous solution, which one has the smallest pK_b value ?

[JEE(Main)-2014]

(1) $(CH_3)_3N$ (2) $C_6H_5NH_2$ (3) $(CH_3)_2NH$ (4) CH_3NH_2

AB0124

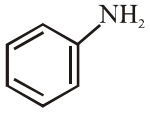
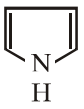
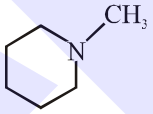
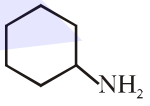
22. Among the following oxoacids, the correct decreasing order of acid strength is : **[JEE(Main)-2014]**

(1) $HClO_4 > HClO_3 > HClO_2 > HOCl$ (2) $HClO_2 > HClO_4 > HClO_3 > HOCl$

(3) $HOCl > HClO_2 > HClO_3 > HClO_4$ (4) $HClO_4 > HOCl > HClO_2 > HClO_3$

AB0125

23. Among the following compounds, the increasing order of their basic strength is:-

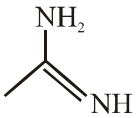
(I)  (II)  (III)  (IV) 

(1) (II) < (I) < (III) < (IV) (2) (I) < (II) < (IV) < (III) **[JEE(Main)-On-Line 2017]**

(3) (II) < (I) < (IV) < (III) (4) (I) < (II) < (III) < (IV)

AB0126

24. The increasing order of basicity of the following compounds is : **[JEE(Main)-2018]**

(a)  (b)  (c)  (d) 

(1) (b) < (a) < (c) < (d) (2) (b) < (a) < (d) < (c)

(3) (d) < (b) < (a) < (c) (4) (a) < (b) < (c) < (d)

AB0127

25. A solution of m-chloroaniline, m-chlorophenol and m-chlorobenzoic acid in ethyl acetate was extracted initially with a saturated solution of $NaHCO_3$ to give fraction A. The left over organic phase was extracted with dilute NaOH solution to give fraction B. The final organic layer was labelled as fraction C. Fractions A, B and C, contain respectively : **[JEE(Main)-2020]**

(1) m-chlorobenzoic acid, m-chloroaniline and m-chlorophenol

(2) m-chloroaniline, m-chlorobenzoic acid and m-chlorophenol

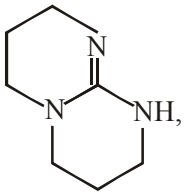
(3) m-chlorobenzoic acid, m-chlorophenol and m-chloroaniline

(4) m-chlorophenol, m-chlorobenzoic acid and m-chloroaniline

AB0154

node06808C-BDXcta\JEE\Advanced\Nurture\Chem\Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength\01_Theory & Ex_p65

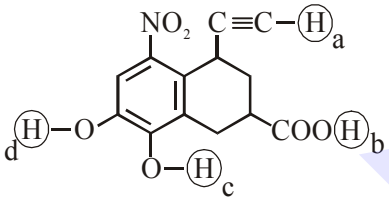
26. The increasing order of pK_b for the following compounds will be : **[JEE(Main)-2020]**

(A) $NH_2 - CH = NH$, (B)  (C) CH_3NHCH_3

(1) (A) < (B) < (C) (2) (C) < (A) < (B) (3) (B) < (A) < (C) (4) (B) < (C) < (A)

AB0155

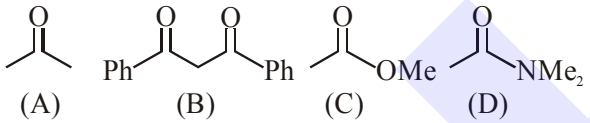
27. Arrange the following labelled hydrogens in decreasing order of acidity : **[JEE(Main)-2020]**



(1) b > c > d > a (2) c > b > a > d (3) b > a > c > d (4) c > b > d > a

AB0156

28. The increasing order of the acidity of the α-hydrogen of the following compounds is :



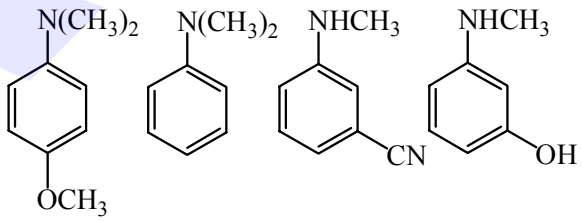
(A) (B) (C) (D) **[JEE(Main)-2020]**

(1) (C) < (A) < (B) < (D) (2) (B) < (C) < (A) < (D)

(3) (A) < (C) < (D) < (B) (4) (D) < (C) < (A) < (B)

AB0157

29. The increasing order of pK_b values of the following compounds is - **[JEE(Main)-2020]**



I II III IV

(1) I < II < IV < III (2) II < IV < III < I

(3) II < I < III < IV (4) I < II < III < IV

AB0158

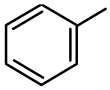
node06808C-BDVKota\JEE[Advanced]\Nurture\Chem Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &B Strength\01_Theory & Ex.p65

30. Which of the following is least basic ? **[JEE(Main)-2021]**

(1) $(CH_3CO)\ddot{N}HC_2H_5$ (2) $(C_2H_5)_3\ddot{N}$ (3) $(CH_3CO)_2\ddot{N}H$ (4) $(C_2H_5)_2\ddot{N}H$

AB0159

31. Which among the following is the strongest acid ? **[JEE(Main)-2021]**

(1) $CH_3CH_2CH_2CH_3$ (2)  (3)  (4) 

AB0160

32. Given below are two statements : **[JEE(Main)-2021]**

Statement I : Aniline is less basic than acetamide.

Statement II : In aniline, the lone pair of electrons on nitrogen atom is delocalised over benzene ring due to resonance and hence less available to a proton.

Choose the most appropriate option ;

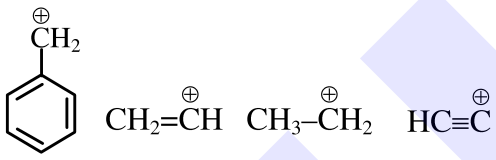
(1) Statement I is true but statement II is false.

(2) Statement I is false but statement II is true.

(3) Both statement I and statement II are true.

(4) Both statement I and statement II are false.

AB0161

33.  **[JEE(Main)-2021]**

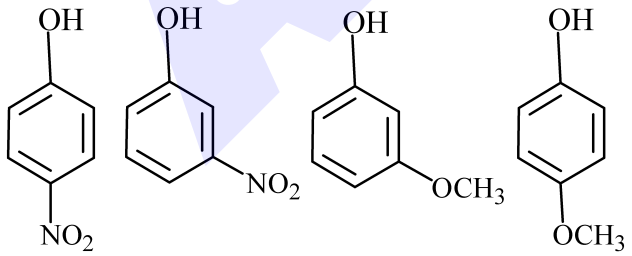
A B C D

The correct order of stability of given carbocation is :

(1) A > C > B > D (2) D >B > C > A (3) D > B > A > C (4) C > A > D > B

AB0162

34. Arrange the following in decreasing acidic strength. **[JEE(Main)-2022]**



(A) (B) (C) (D)

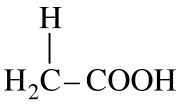
(1) A > B > C > D (2) B > A > C > D

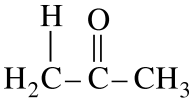
(3) D > C > A > B (4) D > C > B > A

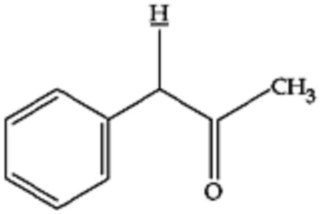
AB0170

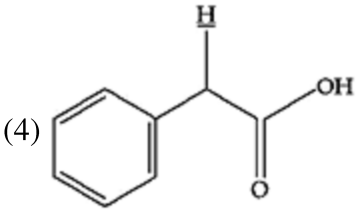
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35. Among the following marked proton of which compound shows lowest pK_a value ? **[JEE(Main)-2022]**

(1) 

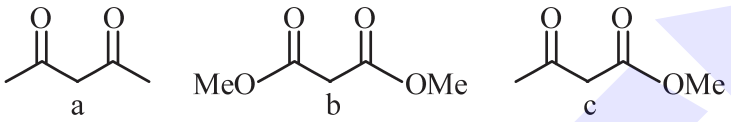
(2) 

(3) 

(4) 

AB0171

36. Which will undergo deprotonation most readily in basic medium? **[JEE(Main)-2023]**



(1) a only (2) c only (3) Both a and c (4) b only

AB0172

37. The increasing order of pKa for the following phenols is **[JEE(Main)-2023]**

(A) 2, 4-Dinitrophenol (B) 4 - Nitrophenol (C) 2, 4, 5- Trimethylphenol

(D) Phenol (E) 3-Chlorophenol

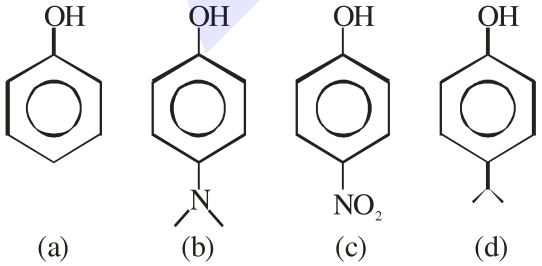
Choose the correct answer from the option given below :

(1) (A), (E), (B), (D), (C) (2) (C), (D), (E), (B), (A)

(3) (C), (E), (D), (B), (A) (4) (A), (B), (E), (D), (C)

AB0173

38. The correct order of pKa values for the following compounds is: **[JEE(Main)-2023]**

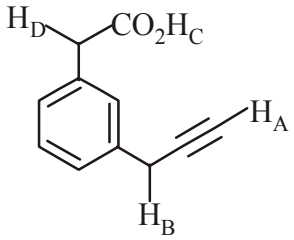


(a) (b) (c) (d)

(1) c > a > d > b (2) b > d > a > c (3) b > a > d > c (4) a > b > c > d

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39. What is the correct order of acidity of the protons marked A-D in the given compounds ?



[JEE(Main)-2023]

(1) $H_C > H_D > H_B > H_A$

(2) $H_C > H_D > H_A > H_B$

(3) $H_D > H_C > H_B > H_A$

(4) $H_C > H_A > H_D > H_B$

AB0175

40. Match List I with List II **[JEE(Main)-2023]**

	List I (Amines)		List II (pK_b)
A.	Aniline	I.	3.25
B.	Ethanamine	II.	3.00
C.	N-Ethylethanamine	III.	9.38
D.	N, N-Diethylethanamine	IV.	3.29

Choose the correct answer from the options given below :-

(1) A-I, B-IV, C-II, D-III

(2) A-III, B-II, C-I, D-IV

(3) A-III, B-II, C-IV, D-I

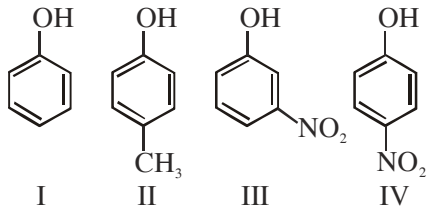
(4) A-III, B-IV, C-II, D-I

AB0176

node06808C-BDXcata\JEE\Advanced\Nurture\Chem\Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength\01_Theory & Ex_p65

EXERCISE # J-ADVANCED

1. In the following compounds **[IIT-JEE-1996]**



The order of acidity is -

(A) III>IV>I>II (B) I>IV>III>II (C) II>I>III>IV (D) IV>III>I>II

AB0137

2. Although phenoxide ion has more number of resonating structures than benzoate ion, benzoic acid is a stronger acid than phenol. Why ? **[IIT-JEE-1997]**

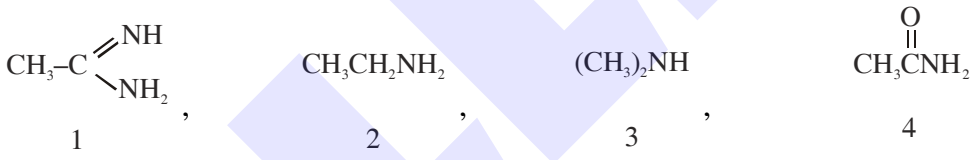
AB0138

3. Amongst the following, the most basic compound is - **[IIT-JEE-2000]**

(A) $C_6H_5NH_2$ (B) $p–NO_2–C_6H_4NH_2$ (C) $m–NO_2–C_6H_4NH_2$ (D) $C_6H_5CH_2NH_2$

AB0139

4. The correct order of basicities of the following compounds is : **[IIT-JEE-2001]**



(A) 2 > 1 > 3 > 4 (B) 1 > 3 > 2 > 4 (C) 3 > 1 > 2 > 4 (D) 1 > 2 > 3 > 4

AB0140

5. **Statement-I :** p-Hydroxybenzoic acid has a lower boiling point that o-hydroxybenzoic acid.

Because

Statement-II : o-Hydroxybenzoic acid has intramolecular hydrogen bonding. **[IIT-JEE-2003]**

(A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I

(B) Statement-I is True, Statement-II is True; Statement-II is NOT a correct explanation for Statement-I

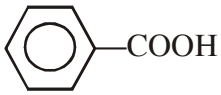
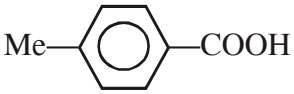
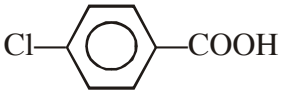
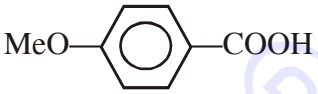
(C) Statement-I is True, Statement-II is False.

(D) Statement-I is False, Statement-II is True.

AB0141

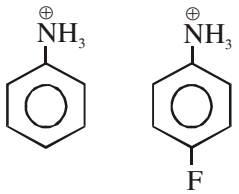
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6. Match K_a values with suitable acid : **[IIT-JEE-2003]**

K_a	Acid
(A) 3.3×10^{-5}	(p) 
(B) 4.2×10^{-5}	(q) 
(C) 6.3×10^{-5}	(r) 
(D) 6.4×10^{-5}	(s) 
(E) 30.6×10^{-5}	(t) 

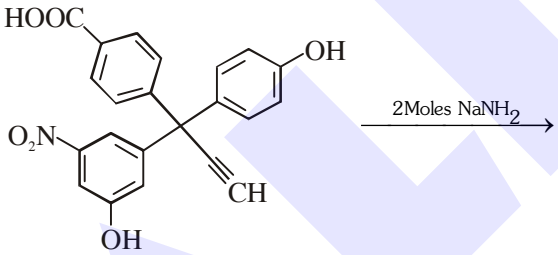
AB0142

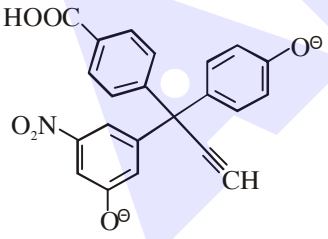
7. (a) Which of the following is more acidic and why ? **[IIT-JEE-2004]**

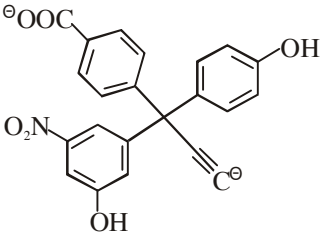


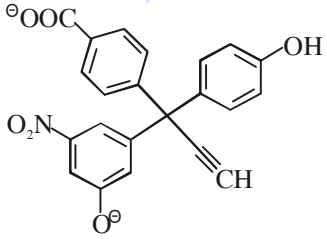
(I) (II)

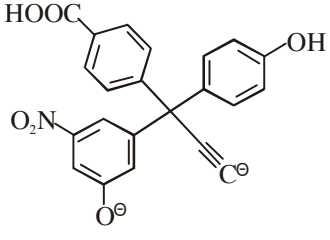
AB0143

8.  A. The product (A) will be : **[IIT-JEE-2007]**

(A) 

(B) 

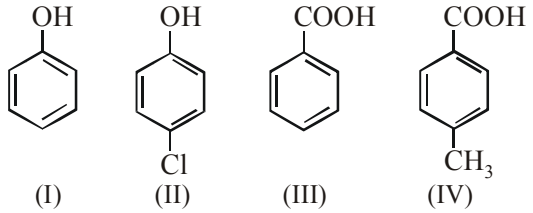
(C) 

(D) 

AB0144

node06808C-BDX\data\JEE\Advanced\Nurture\Chem\Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength\01_Theory & Ex_p65

9. The correct acidity order of the following : **[IIT-JEE-2009]**



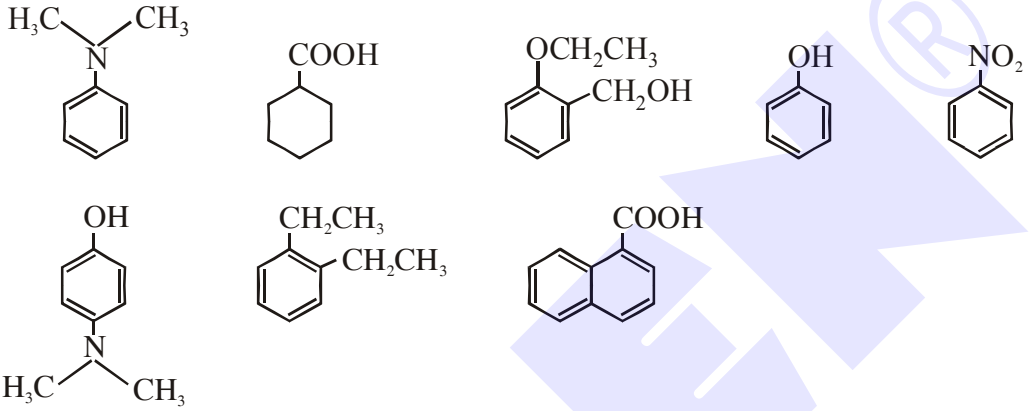
(I) (II) (III) (IV)

(A) (III) > (IV) > (II) > (I) (B) (IV) > (III) > (I) > (II)

(C) (III) > (II) > (I) > (IV) (D) (II) > (III) > (IV) > (I)

AB0145

10. Amongst the following, the total number of compounds soluble in aquesous NaOH is: **[IIT-JEE-2010]**



AB0146

11. Among the following compounds, the most acidic is **[IIT-JEE-2011]**

(A) p-nitrophenol (B) p-hydroxybenzoic acid

(C) o-hydroxybenzoic acid (D) p-toluic acid

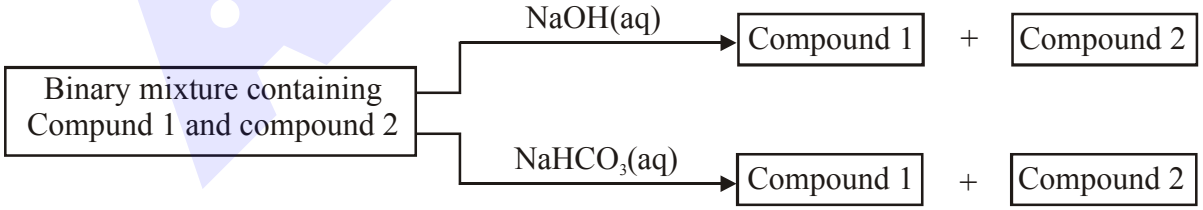
AB0147

12. The carboxyl functional group (–COOH) is present in - **[IIT-JEE-2012]**

(A) picric acid (B) barbituric acid (C) ascorbic acid (D) aspirin

AB0148

13. Identify the binary mixtures (s) that can be separated into the individual compounds, by differential extraction, as shown in the given scheme - **[IIT-JEE-2012]**



(A) C_6H_5OH and C_6H_5COOH (B) C_6H_5COOH and $C_6H_5CH_2OH$

(C) $C_6H_5CH_2OH$ and C_6H_5OH (D) $C_6H_5CH_2OH$ and $C_6H_5CH_2COOH$

AB0149

14. The compound that does NOT liberate CO_2, on treatment with aqueous sodium bicarbonate solution, is - **[JEE-ADVANCED-2013]**

(A) Benzoic acid (B) Benzenesulphonic acid (C) Salicylic acid (D) Carbolic acid (phenol)

AB0150

node06808C-BDVKota\JEE(Advanced)\Nurture\Chem Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &B Strength\01_Theory & Ex.p65

15. Hydrogen bonding plays a central role in the following phenomena **[JEE-ADVANCED-2014]**

(A) Ice floats in water

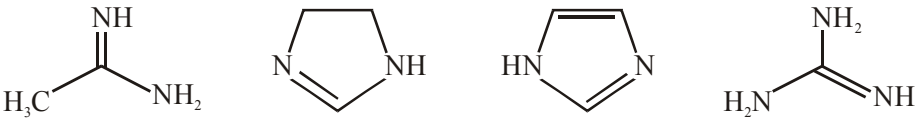
(B) Higher Lewis basicity of primary amines than tertiary amines in aqueous solutions

(C) Formic acid is more acidic than acetic acid

(D) Dimerisation of acetic acid in benzene

AB0151

16. The order of basicity among the following compounds is **[JEE-ADVANCED-2017]**

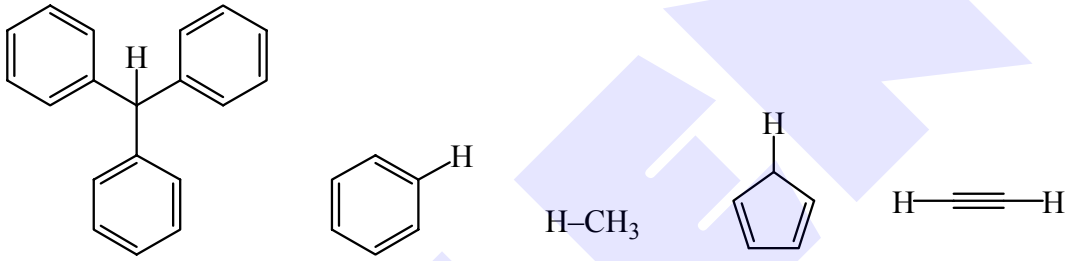


I II III IV

(A) II > I > IV > III (B) IV > II > III > I (C) I > IV > III > II (D) IV > I > II > III

AB0152

17. With respect to the compounds **I-V**, choose the correct statement(s). **[IIT-2020]**



I II III IV V

(A) The acidity of compound **I** is due to delocalization in the conjugate base.

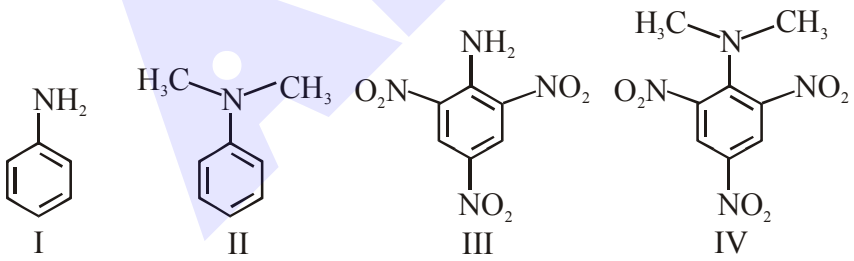
(B) The conjugate base of compound **IV** is aromatic.

(C) Compound **II** becomes more acidic, when it has a -NO_2 substituent.

(D) The acidity of compounds follows the order **I > IV > V > II > III.**

AB0163

18. Consider the following four compounds **I**, **II**, **III**, and **IV**. **[IIT-2020]**



I II III IV

Choose the correct statement(s).

(A) The order of basicity is **II** > **I** > **III** > **IV**.

(B) The magnitude of pK_b difference between **I** and **II** is more than that between **III** and **IV**.

(C) Resonance effect is more in **III** than in **IV**.

(D) Steric effect makes compound **IV** more basic than **III**.

AB0164

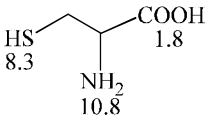
node06808C-BDXtab\JEE\Advanced\Nurture\Chem\Module\3-Nomen-EDE-A &B Strength-Isom-Purification\Eng\03_A &-B Strength\01_Theory & Ex_p65

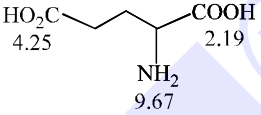
ANSWER-KEY

EXERCISE # O-I

1. (i) a > b > c > d, (ii) a > b > c , (iii) c > b > a, (iv) a > b > c,
(v) c > b > a, (vi) a > b > c (vii) d > c > b > a , (viii)d > c > b > a,
(ix) d > b > a > c, (x) d > a > c> b

2. (a) 2 ; (b) 2 ; (c) 1 ; (d) 1 **3.** (a) 2 ; (b) 2 ; (c) 2 **4.** (C) **5.** (B)
6. (B) **7.** (B) **8.** (A) **9.** (A)

10. (i) cysteine : 

(ii) glutamic acid : 

11. (a) 3 < 2 < 1 ; (b) 1 < 2 < 3 ; (c) 3 < 2 < 1 ; (d) 2 < 1 < 3 ; (e) 2 < 3 < 1

12. (i) d > c > a > b, (ii) a > b > c, (iii) c > a > b > d, (iv) d > b > c > a,
(v) a > b > c, (vi) b > a (vii) c > a > b

13. (i) b , (ii) a, (iii) b , (iv) b **14.** (C) **15.** (C) **16.** (B)
17. (C) **18.** (B) **19.** (4) **20** (C)

EXERCISE # O-II

1. (i) a > b> c> d, (ii) a > b > c > d, (iii) c > b> d> a , (iv) d > c > b > a
(v) a > b > c, (vi) c > b > a (vii) c > a > b, (viii) b > c > a, (ix) c > d > b > a

2. (i) a > b > c, (ii) d > c > b > a, (iii) b > c > a, (iv) d > c > b > a,
(v) b > a > c, (vi) b > a, (vii) c > b > a, (viii) d > a > b > c
(ix) d > c > b > a

3. (i) d, (ii) b, (iii) a, (iv) a

4. (i) b > a > d > c, (ii) b > a > c > d, (iii) a > b > c > d

5. (A) **6.** (A) **7.** (A) **8.** (A)

9. (a) i , (b) ii , (c) i , (d) ii **10.** (a) 2 ; (b) 1 ; (c) 1 ; (d) 1 ; (e) 3

11. (a) 2; (b) 1;(c) 2; (d) 2 **12.** (a) 1 > 2 > 3 ; (b) 1 < 2 < 3 ; (c) 3 < 1 < 2 ; (d) 2 < 1 < 3

13. (a) 2 < 1 < 3 ; (b) 1 < 2 < 3 **14.** (C) **15.** (B)

16. 3, 3 basic groups are $\ddot{N}H_2$; –NH–; O^- **17.** (6)

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EXERCISE # S-I

1. (C) **2. (C)**

3. (A) – R, S, T ; (B) – P, R, S, T ; (C) – P, Q, R, S, T ; (D) – P, Q, R, S, T

4. (i) , (ii) (iii) (iv), (v) (ix) **5. (D)** **6. (A)**

7. (A) – P,T ; (B) – R; (C) – Q ; (D) – S, T **8. (A) – R ; (B) – S; (C) – P ; (D) – Q, T**

9. (A) **10. (A)** **11. (A, D)** **12. (D)**

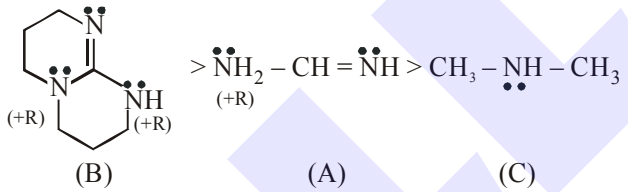
13. (A, B, C, D) **14. (B)** **15. (C)**

EXERCISE # JEE-MAIN

1. (3)	**2. (4)**	**3. (3)**	**4. (4)**	**5. (4)**
6. (3)	**7. (4)**	**8. (3)**	**9. (1)**	**10. (2)**
11. (3)	**12. (1)**	**13. (1)**	**14. (2)**	**15. (3)**
16. (2)	**17. (4)**	**18. (3)**	**19. (2)**	**20. (3)**
21. (3)	**22. (1)**	**23. (3)**	**24. (2)**	**25. (3)**

26. (3)

Sol. Base strength order



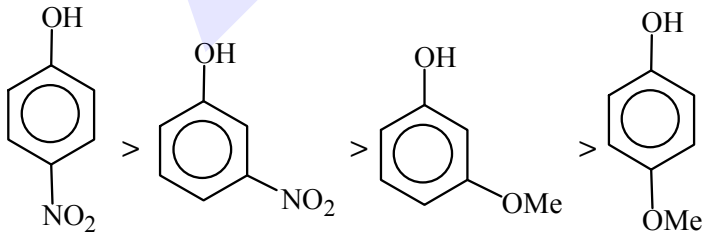
(B) (A) (C)

pk_b order (C > A > B)

27. (1)	**28. (4)**	**29. (1)**	**30. (3)**	**31. (4)**
32. (2)	**33. (1)**			

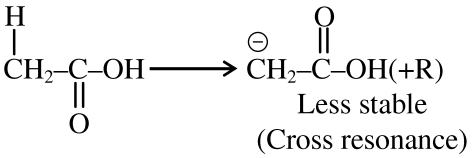
34. (1)

Sol. The correct order of acid strength is

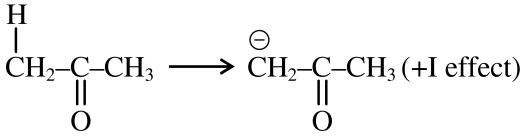


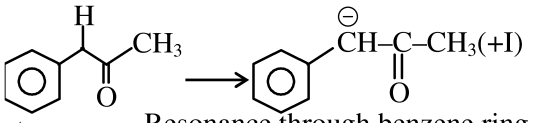
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35. (3)

Sol. (A) 

Less stable
(Cross resonance)

(B) 

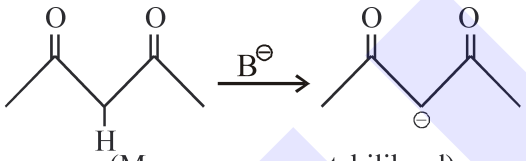
(C) 

(most acidic)
Resonance through benzene ring (most stable)

So it has least pKa value.

36. (1)

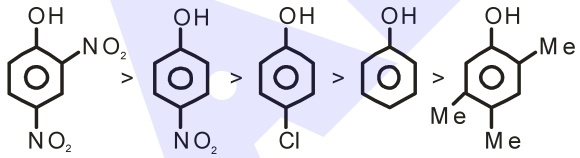
Sol. Most easily deprotonation



(More resonance stabilibsed)

37. (4)

Sol. Order of acidity for following phenol is



- M and - I increases acidity

+ M and + I decreases acidity

38. (2)

Sol. Due to -M effect of -NO2 group, it increases acidity

+M effect of N (CH3)2 decreases acidity.

Hyperconjugation of isopropyl decrease acidity

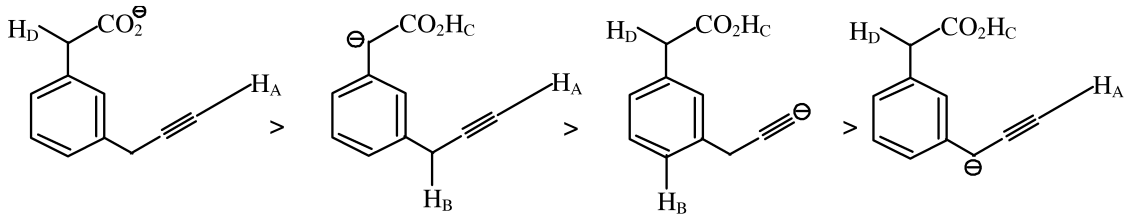
order of acidic strength

(c) > (a) > (d) > (b)

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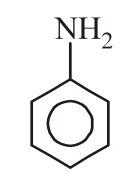
39. (2)

Sol. acidity of an acid depends upon the stability of its conjugate base

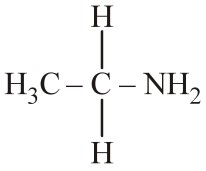


40. Allen Ans. (4)

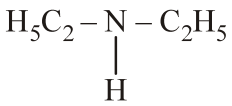
Sol.



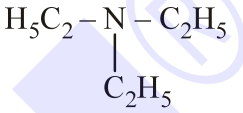
(A) Aniline



(B) Ethanamine



(C) N-Ethylethanamine



(D) N, N-Diethylethanamine

$$\text{Basic Strength } \alpha \frac{1}{pK_b}$$

Order for pKb : A > B > D > C

EXERCISE # J-ADVANCED

1. (D) **2. Benzoate has equivalent resonating structures** **3. (D)**

4. (B) **5. (D)** **6. A-(s) ; B-(q) ; C-(p) ; D-(r) ; E-(t)**

7. (II is most acidic) **8. (C)** **9. (A)** **10. (4)** **11. (C)**

12.(D) **13.(B, D)** **14.(D)** **15.(A, B, D)**

16. (D) **17. (A,B,C)** **18. (C,D)**

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Important Notes

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